Ejercicios de tcpdump

🔷 **Ejercicios básicos**

1. **Listar interfaces disponibles**

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tcpdump -D

```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump -D
1.eth0 [Up, Running, Connected]
2.any (Pseudo-device that captures on all interfaces) [Up, Run
ning]
3.lo [Up, Running, Loopback]
4.bluetooth-monitor (Bluetooth Linux Monitor) [Wireless]
5.nflog (Linux netfilter log (NFLOG) interface) [none]
6.nfqueue (Linux netfilter queue (NFQUEUE) interface) [none]
7.dbus-system (D-Bus system bus) [none]
8.dbus-session (D-Bus session bus) [none]
```

👉 Muestra todas las interfaces de red que tcpdump puede usar.

2. **Capturar tráfico en la interfaz eth0**

---

tcpdump -i eth0

```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump -i eth0
tcpdump: verbose output suppressed, use -v[v]... for full prot
ocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot lengt
h 262144 bytes
11:43:48.614821 IP 192.168.0.21.54174 > 239.255.255.250.1900:
UDP, length 324
11:43:48.646534 IP 192.168.0.21.51753 > 239.255.255.250.1900:
UDP, length 380
11:43:48.676602 IP 192.168.0.21.33418 > 239.255.255.250.1900:
UDP, length 333
```

👉 Ver los primeros 20 paquetes que pasan por eth0.

3. **Guardar captura en un fichero**

tcpdump -i eth0 -w trafico.log

```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump -i eth0 -w trafico.log
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snaps
hot length 262144 bytes
^C548 packets captured
585 packets received by filter
0 packets dropped by kernel

┌──(kali㉿kali)-[~]
└─$ ls
Descargas   Escritorio  Música      Público     Vídeos
Documentos  Imágenes    Plantillas  trafico.log
```

👉 Luego ábrelo con:

tcpdump -r trafico.log

```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump -i eth0 -r trafico.log
reading from file trafico.log, link-type EN10MB (Ethernet), sn
apshot length 262144
11:44:48.631935 ARP, Request who-has 192.168.0.59 (Broadcast)
tell 192.168.0.90, length 46
11:44:48.653946 ARP, Request who-has 192.168.0.60 (Broadcast)
tell 192.168.0.90, length 46
11:44:48.662941 ARP, Request who-has 192.168.0.61 (Broadcast)
tell 192.168.0.90, length 46
11:44:48.712376 ARP, Request who-has 192.168.0.62 (Broadcast)
tell 192.168.0.90, length 46
```

🔷 **Filtrado por protocolos y puertos**

4. **Capturar solo tráfico TCP en eth0**

tcpdump -i eth0 tcp

```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump -i eth0 tcp
tcpdump: verbose output suppressed, use -v[v]... for full prot
ocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot lengt
h 262144 bytes
11:48:47.481400 IP 192.168.0.76.58926 > 216.72.190.35.bc.googl
eusercontent.com.https: Flags [S], seq 1808517698, win 64240,
options [mss 1460,sackOK,TS val 2751446616 ecr 0,nop,wscale 7]
, length 0
11:48:47.496355 IP 216.72.190.35.bc.googleusercontent.com.http
s > 192.168.0.76.58926: Flags [S.], seq 404177123, ack 1808517
699, win 65535, options [mss 1412,sackOK,TS val 3744107961 ecr
 2751446616,nop,wscale 8], length 0
11:48:47.496378 IP 192.168.0.76.58926 > 216.72.190.35.bc.googl
```

5. **Capturar tráfico en el puerto 80 (HTTP)**

tcpdump -i eth0 tcp port 80

```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump -i eth0 tcp port 80
tcpdump: verbose output suppressed, use -v[v]... for full prot
ocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot lengt
h 262144 bytes
^C
0 packets captured
0 packets received by filter
0 packets dropped by kernel
```

6. **Capturar tráfico DNS (UDP puerto 53)**

tcpdump -i eth0 udp port 53

```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump -i eth0 udp port 53
tcpdump: verbose output suppressed, use -v[v]... for full pr
ocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot len
h 262144 bytes
```

🔷 **Filtrado por IP**

7. **Tráfico desde una IP concreta (host origen)**

tcpdump -i eth0 src 192.168.0.76

```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump -i eth0 src 192.168.0.76
tcpdump: verbose output suppressed, use -v[v]... for full prot
ocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot lengt
h 262144 bytes
^C
0 packets captured
0 packets received by filter
0 packets dropped by kernel
```

8. **Tráfico hacia una IP concreta (host destino)**

tcpdump -i eth0 dst 192.168.0.76

```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump -i eth0 dst 192.168.0.76
tcpdump: verbose output suppressed, use -v[v]... for full prot
ocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot lengt
h 262144 bytes
12:10:56.612230 IP 93.243.107.34.bc.googleusercontent.com.http
s > 192.168.0.76.45022: Flags [P.], seq 874047154:874047193, a
ck 2820644492, win 1045, options [nop,nop,TS val 3498188564 ec
r 2552695262], length 39
12:10:56.687502 IP 254.red-80-58-61.staticip.rima-tde.net.doma
in > 192.168.0.76.50806: 17037 NXDomain 0/1/0 (120)
12:10:56.709276 IP 254.red-80-58-61.staticip.rima-tde.net.doma
```

9. **Tráfico de y hacia una IP concreta (host completo)**

tcpdump host 192.168.0.76

```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump host  192.168.0.76
tcpdump: verbose output suppressed, use -v[v]... for full pro
ocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot leng
h 262144 bytes
^C
0 packets captured
0 packets received by filter
0 packets dropped by kernel
```

🔷 Filtrado por MAC y red

10. **Tráfico con destino a una dirección MAC**

tcpdump ether dst 08:00:27:6f:98:bb

ip link para ver la mac

```
┌──(kali㉿kali)-[~]
└─$ ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UN
KNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_c
odel state UP mode DEFAULT group default qlen 1000
    link/ether 08:00:27:6f:98:bb brd ff:ff:ff:ff:ff:ff
```

```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump ether dst 08:00:27:6f:98:bb
tcpdump: verbose output suppressed, use -v[v]... for full prot
ocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot lengt
h 262144 bytes
^C
0 packets captured
0 packets received by filter
0 packets dropped by kernel
```

11. **Tráfico de una red completa**

tcpdump net 192.168.0.0/24

```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump net 192.168.0.0/24
tcpdump: verbose output suppressed, use -v[v]... for full prot
ocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot lengt
h 262144 bytes
11:58:56.043896 ARP, Request who-has 192.168.0.1 tell 192.168.
0.76, length 28
11:58:56.044204 ARP, Reply 192.168.0.1 is-at 00:a0:26:d2:68:9a
 (oui Unknown), length 46
11:58:56.252788 IP 192.168.0.76.41699 > 254.red-80-58-61.stati
cip.rima-tde.net.domain: 47345+ PTR? 1.0.168.192.in-addr.arpa.
 (42)
```

🔷 Visualización de datos

12. **Mostrar el contenido en ASCII**

tcpdump -i eth0 -A

```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump -i eth0 -A
tcpdump: verbose output suppressed, use -v[v]... for full prot
ocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot lengt
h 262144 bytes
11:59:53.308582 ARP, Request who-has 10.0.2.2 tell 10.0.2.15,
length 46
..........'.C.
.........
.....................
11:59:53.318419 IP 192.168.0.38.mdns > mdns.mcast.net.mdns: 0
PTR (QM)? _spotify-connect._tcp.local. (45)
E..I.6@........&.........5..............._spotify-connect._tcp
.local.....
```

13. **Mostrar el contenido en hexadecimal**

tcpdump -i eth0 -XX

```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump -i eth0 -XX
tcpdump: verbose output suppressed, use -v[v]... for full prot
ocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot lengt
h 262144 bytes
12:00:26.294606 IP 192.168.0.38.57621 > 192.168.1.255.57621: U
DP, length 40
        0x0000:  ffff ffff ffff 96d4 9398 92f0 0800 4500  ....
..........E.
        0x0010:  0044 0600 4000 4011 b133 c0a8 0026 c0a8  .D..
@.@..3...&..
        0x0020:  01ff e115 e115 0030 485d 5370 6f74 5564  ....
...0H]SpotUd
        0x0030:  7030 dade 4dc8 561e 3b2e 0001 0000 5fb4  p0..
```

Ejercicios combinados (operadores lógicos)

14. Tráfico desde una IP y en puerto 80

tcpdump -i eth0 src 192.168.0.76 and tcp port 80

```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump -i eth0 src 192.168.0.48 and tcp port 80
tcpdump: verbose output suppressed, use -v[v]... for full prot
ocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot lengt
h 262144 bytes
^C
0 packets captured
0 packets received by filter
0 packets dropped by kernel
```

15. Tráfico desde una IP excluyendo UDP

tcpdump -i eth0 src 192.168.0.76 and not udp

```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump -i eth0 src 192.168.0.48 and not udp
tcpdump: verbose output suppressed, use -v[v]... for full prot
ocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot lengt
h 262144 bytes
12:03:14.218479 ARP, Request who-has 192.168.0.94 tell 192.168
.0.48, length 46
^C
1 packet captured
1 packet received by filter
0 packets dropped by kernel
```

16. Tráfico desde una IP en HTTP o HTTPS

tcpdump -i eth0 src 192.168.0.76 and (port http or https)

```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump -i eth0 src 192.168.0.76 and port https
tcpdump: verbose output suppressed, use -v[v]... for full prot
ocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot lengt
h 262144 bytes
12:05:00.526591 IP 192.168.0.76.47792 > mad07s22-in-f3.1e100.n
et.https: Flags [P.], seq 4173280014:4173280053, ack 304481583
3, win 891, options [nop,nop,TS val 693990946 ecr 2056199985],
 length 39
```

👉 Con esta lista de **16 ejercicios** tienes un recorrido completo:

- Empezando por capturar y guardar tráfico.
- Luego filtrando por protocolo, puerto, IP, MAC y red.
- Finalmente usando operadores lógicos y visualización en distintos formatos.